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M 012 -SHOP FABRICATED TANKS

1 SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2. The shop fabricated tanks shall include but not be limited to the tanks specified in Clause 3 herein and its required electrical, instrumentation and control equipment specified in Clauses 7 and 8 herein.

Bidder shall add all required tanks to provide complete and functional systems. Shop fabricated tanks specified elsewhere in the Specifications shall also be subject to the requirements of this specification where applicable.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

3 OPERATING AND DESIGN CONDITIONS

Minimum requirements are specified in the Table below. The Bidder shall ensure that all of these requirements are satisfied in the design of the tanks.

3.1 Shop fabricated Tanks

Shop fabricated tanks Description		Tank 1	Tank 2	Tank 3	Tank 4	Tank 5
1.	Name of Tank	Clean Turbine Oil Tank	Dirty Turbine Oil Tank	Phosphate Tank	Ammonium Hydroxide Solution Tank	Carbohydrazide Solution Tank
2.	Specification No.	M 209	M 209	M-405	M-405	M-405
3.	Location	Steam Turbine Building	Steam Turbine Building	Chemical Injection Station	Chemical Injection Station	Chemical Injection Station
4.	Quantity	1	1	1	1	1
5.	Fluid contained	Turbine Oil	Turbine Oil	Phosphate	Ammonium hydroxide	Carbohydrazide Solution
6.	Capacity (1)	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO
7.	Design pressure (bar)	Atmospheric	Atmospheric	Atmospheric	Atmospheric	Atmospheric
8.	Design temp. (°C)	120	120	50	50	50
9.	Material	Carbon Steel	Carbon Steel	-C/D -ABO	-C/D -ABO	-C/D -ABO
10.	Governing code	API 650	API 650	AWWA D100	AWWA D100	AWWA D100
11.	Minimum Accessories *	7,10,11	7,10,11	1,12	1,12	1,12
12.	Minimum connections **	A,B,C,D,F	A,B,C,D,F	A,B,H	A,B,D,H	A,B,H
13.	Interior coating	No	No	-C/D -ABO	-C/D -ABO	-C/D -ABO

C/D = Bidder to determine
ABO = Approval by the Owner

Shop fabricated tanks Description		Tank 1	Tank 2	Tank 3	Tank 4	Tank 5
14.	Minimum Testing requirement ***	LT,LP	LT,LP	LT,LP	LT,LP	LT,LP
15.	Specification No.	- C/D - ABO	- C/D - ABO	- C/D - ABO	- C/D - ABO	- C/D - ABO
16.	Ladder	- C/D - ABO	- C/D - ABO	- C/D - ABO	- C/D - ABO	- C/D - ABO

C/D = Bidder
ABO = Approval by the Owner

Shop fabricated tanks Description		Tank 6	Tank 7	Tank 8	Tank 9
1.	Name of tank	Blowdown Tank (Vertical Tank)	Atmospheric start up flash tank vertical tank	Demineralized Water Caustic Hot Water	Demineralized Water Acid Storage Tank
2.	Specification	M 301	M 301	M 401	M 401
3.	Location	HRSG area	HRSG area	Inside water Treatment Plant Building	Inside of Water Treatment Building
4.	Quantity	1	1	1	1
5.	Fluid Contained	Steam/Condensate	Steam/Condensate	Water	Sulphuric Acid
6.	Capacity	-C/D -ABO	-C/D -ABO	-C/D -ABO	Minimum 2 minute Storage at demineralized design flow.
7.	Design pressure (bar)	-C/D -ABO -C/D -ABO	-C/D -ABO -C/D -ABO	-C/D -ABO	-C/D -ABO
8.	Design Temp. (°C)	-ABO	-ABO	100 ⁰ (minimum)	40 ⁰ (minimum)
9.	Material	Carbon steel	Carbon steel	Carbon steel	-C/D -ABO
10.	Governing code	ASME boiler & Pressure code Section VIII	ASME boiler & Pressure Code Section VIII	ASME Sec. VIII	AWWA D100

C/D = Bidder to determine
ABO = Approval by the Owner

Shop fabricated tanks Description		Tank 6	Tank 7	Tank 8	Tank 9
11.	Minimum Accessories*	1,8,12,13, 14,15	1,8,9,12, 13,14	5,9,12	1,2,12
12.	Minimum Connections**	A,B,C,D,F,G	A,B,C,D,F,H	A,B,C,G	A,B,C,D,E,F
13.	Interior coating	No	No	-C/D -ABO	-C/D -ABO
14.	Minimum testing Requirement ***	XR,LT,LP	XR,LT,LP	LT,XR,LP	LT,XR,LP
15.	Platform	No	No	-C/D -ABO	-C/D -ABO
16.	Ladder	No	No	-C/D -ABO	-C/D -ABO
17.	Insulation	Yes	Yes	Yes	-

C/D = Bidder to determine
ABO = Approval by the Owner

Shop fabrication tanks Description		Tank 10	Tank 11	Tank 12	Tank 13	Tank 14	Tank 15
1.	Name of Tank	Demineralized Water Caustic Storage Tank	Waste Water Treatment Acid Storage Tank	Waste Water Treat- ment Caustic Day Tank	Waste Water Treatment Acid Day Tank	Potable Water Head Tank (Horizontal)	Closed Aux Cooling Water Head Tank (Horizontal)
2.	Specification No.	M-401	M-404	M-404	M-404	M-403/S-420	M-402
3.	Location	Water Treatment Area	Chemical Cleaning Waste Water Treat- ment Plant	Outside Water Treatment Plant	Outside Water Treatment Plant	Administration Building & Main Control Building	Steam Turbine Building
4.	Quantity	1	1	1	1	1	1
5.	Fluid Contained	Caustic	Acid	Caustic	HCL	Potable Water	Demineralized Water
6.	Capacity	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO
7.	Design pressure (bar)	Atmospheric	Atmospheric	Atmospheric	Atmospheric	Atmospheric	Atmospheric
8.	Design temp. (°C)	40 ⁰ (minimum)	40 ⁰ (minimum)	40 ⁰ (minimum)	40 ⁰ (minimum)	40 ⁰ (minimum)	40 ⁰ (minimum)
9.	Material	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	Carbon Steel	Carbon Steel
10.	Governing code	AWWA D100	AWWA D100	AWWA D100	AWWA D100	AWWA D100	AWWA D100

PART 4 Technical Requirements
SECTION 3 Technical Specifications
M-012 Shop Fabricated Tanks

Shop fabrication tanks Description		Tank 10	Tank 11	Tank 12	Tank 13	Tank 14	Tank 15
11.	Minimum Accessories *	1,2,12	1,2,3,12	1,12	1,12	1,14	1,14
12.	Minimum connections **	A,B,C,D,E,F	A,B,C,D,E,F	A,B,D,E,F	A,B,C,D,E,F	A,B,D,E,F	A,B,C,E,F,H
13.	Interior coating	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	Yes	Yes
14.	Minimum Testing Requirement ***	LT, XR, LP	LT, XR, LP	LT, XR, LP	LT, XR, LP	LT, LP	LT, LP
15.	Platform	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO
16.	Ladder	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO	-C/D -ABO

C/D = Bidder to determine
ABO = Approval by the Owner

Shop fabricated tanks Description		Tank 16	Tank 17	Tank 18
1.	Name of Tank	Diesel fire Pump Day Tank	Miscellaneous Drain Tank	Diesel Emergency Generator Tank
2.	Specification No.		NA	
3.	Location	Raw Water Storage Tank Area	Power house	Power house
4.	Quantity	1	1	1
5.	Fluid contained	Fuel Oil	Condensate Water	HSD Oil
6.	Capacity (l)	1000L	-C/D -ABO	1000L
7.	Design pressure (kg/cm2)	Atmospheric (minimum)	Atmospheric (minimum)	Atmospheric (minimum)
8.	Design temperature (0C)	Ambient	80 (maximum)	Ambient
9.	Material	Carbon Steel	Carbon Steel	Carbon Steel
10.	Governing code	ASME/API	ASME/API	ASME/API
11.	Minimum Accessories *	3,12	1,5,9,12,13 and 14	3,12,13
12.	Minimum connections **	A,B,C,D,E,H	A,B,C,D,E,F,G,H	A,B,C,D,E,H
13.	Interior coating	No	Yes	No
14.	Minimum Testing Requirements ***	XR,LT,LP	LT, LP	XR, LT, LP
15.	Platform	-C/D -ABO	-C/D -ABO	-C/D -ABO
16.	Ladder	-C/D -ABO	-C/D -ABO	-C/D -ABO

C/D = Bidder to determine
ABO = Approval by the Owne

Minimum Accessories *

1. Gauge Glass
2. Dry Breather
3. Level Indicator (Dial Type)
4. Type S Uehling Tank-O-meter or equivalent with hand pump actuation
5. Temperature Indicator
6. Bubbler type level indicator
7. Explosion-proof level switches and alarm equipment
8. Pressure Indicator
9. Relief valve
10. Flame arrestor
11. Float operated board and cursor type level indicator
12. Level switches
13. Level controller
14. Level measurement stand pipe
15. Temperature controller

Minimum Connections **

- A Inlet
- B Outlet
- C Drains
- D Vents
- E Overflow
- F Connections for Level Measurement Standpipe
- G Connections for Temperature Measurement
- H Connections for Pump Recirculation

Minimum Testing Requirements * * *

- XR Radiography
- LT Leak Testing
- LP Liquid Penetrant Examination

MP Magnetic Particle Examination

3.2 Additional Requirements

Design and construction of the clean and dirty turbine oil tanks shall be based on the recommendations of API Standard 650 (Welded Steel Tanks for Oil Storage), the National Fire Protection Association and acceptable industry practices and shall bear the Underwriters Laboratory label.

4 DETAILED REQUIREMENTS

4.1 Allowable Materials

For all the carbon steel materials listed for plate, pipe, castings, etc., the allowable stresses, for ASME SA-285 shall govern. Adequate corrosion allowance shall be added to the wall thickness of the tanks.

Requirements for tanks to be designed, constructed, fabricated and tested in accordance with the requirements of ASME BPVC Section VIII are indicated underneath an asterisk.

4.1.1 Plate

Carbon Steel : ASTM A285, preferred (A283 and A36 on approval of Owner).

Other than Carbon Steel : Bidder to determine and shall be approved by the Owner.

4.1.2 Pipe

Carbon Steel : ASTM A106, Grade A or B; ASTM A53.

Other than Carbon Steel : Bidder to determine and shall be approved by the Owner.

4.1.3 Castings

Carbon Steel : ASTM A216, ASTM A27.

Other than Carbon Steel : Bidder to determine and shall be approved by the Owner.

4.1.4 Fittings

Carbon Steel : ASTM A105, ASTM A181

Other than Carbon Steel : Bidder to determine and shall be approved by the Owner.

4.1.5 Forgings

Carbon Steel :ASTM A182

Other than Carbon Steel : Bidder to determine and shall be approved by the Owner.

4.1.6 Welding Electrodes

All electrode, filler wire and consumable insert materials shall be selected to match the mechanical and chemical characteristics of the parent metal.

4.1.7 Structural Supports

Carbon Steel: ASTM A36

Carbon Steel: ASME SA-36, SA-285, SA-283, SA-306 or SA-105

4.1.8 Internals

Piping materials shall be as specified above.

4.1.9 Mixed Eductors

Carbon Steel Tanks..... Cast iron or bronze

Other than Carbon Steel: Bidder to determine and shall be approved by the Owner.

Fasteners

U-bolts or other fasteners in carbon steel tanks shall be of carbon steel. In all other tanks U-bolts shall be specified by the Bidder and approved by the Owner.

4.2 Fabrication and Construction

Tanks fabricated from high Alloy steel shall have a minimum 4.7 mm thick wall.

4.2.1 Attachments

Where the tank is fabricated of high alloy steel, a high alloy steel pad of adequate thickness shall be provided between dissimilar materials at points of attachment. If the carbon steel part is to be welded to the tank in the field, the high alloy steel pad shall be welded to the part in the shop so that welds of only similar materials are made in the field.

Tanks shall be self-supporting between supports.

Tanks shall be furnished complete with all required nozzles, vents, drains and flanges, manholes, hand holes and access openings with covers, gaskets and bolting, ladders, platforms, hand-railing, inside piping with suitable supports, if required; and such braces, brackets, legs, etc., as required, including all other appurtenances shown on the drawings.

4.2.2 Material Cutting

Carbon steel may be cut to size or shape by machining, shearing or thermal cutting. Stainless steel or alloy steel may be cut to size or shape by machining or shearing.

Edges shall be free from any defects or distortions which would adversely affect the service ability of the member. All burrs and similar defects shall be carefully removed. The edges of flame-cut steel shall be machined to remove notches, the heat affected zone or both.

4.2.3 Welding

Note: All welding shall be performed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII, with the additional requirement that all welds shall be full-penetration welds, with the exception of structural attachment welds which may be partial penetration (fillet) welds.

All welding shall be performed in accordance with written qualified welding procedure specifications, which are in accordance with the ASME Boiler and Pressure Vessel Code, Section IX.

Welding shall be planned and conducted to minimize warping or distortion.

All welds shall be cleaned of slag flux between passes and following the final deposit.

All double-welded joints shall be back-chipped or arc-gouged to sound metal prior to depositing the second side. Arc gouging is only allowed for ferritic materials.

4.2.4 Welded Joints

All shell joints shall be double-welded, full-penetration butt joints.

All bottom joints in tanks (except flat-bottom tanks) shall be double-welded, full-penetration butt joints. Lap joints may be used for tanks with supported bottoms.

Conical roof-plate joints may be single-welded, full-fillet lap joints. The overlap shall be 5 times the plate thickness but need not exceed 25 mm.

All other roof-plate joints shall be double-welded, full-penetration butt joints.

All nozzle to vessel welds shall be full-penetration welds.

4.3 Additional Requirements for Steel Tanks

4.3.1 Nozzles and Pipe

All internal or external piping and all tank nozzles and nozzles flanges (where required) and all manholes shall be of the same base material as the tank shell, unless otherwise specified.

Nozzles and pipe for carbon steel tanks shall be as follows, except where conditions require heavier wall thickness:

- a. Sizes 50 mm and smaller: Schedule 80
- b. Sizes 80 mm to 250 mm inclusive: Schedule 40

Nozzles and pipe for other than Carbon steel tanks shall be as follows, except where conditions require heavier wall thickness:

- a. Sizes 300 mm and smaller: Schedule 40S
- b. Sizes 350 mm and larger: 3/8-inch wall thickness or metric equivalent
Flanges shall be of the welding neck or slip-on type whichever is better.

Weld-end nozzle terminals for carbon steel and other than Carbon steel tanks shall be as follows:

- a. Sizes 50 mm and smaller: Socket-weld end
- b. Sizes 80 mm NPS and larger: for reference see Part 4 Technical Specification M-001 Piping.

The forces and moments imposed by the nozzles on the tanks shall be considered by the Bidder in his design.

4.3.2 Manholes and Handholes, etc:

The dimensions of the manholes and handholes shall be according to the applicable standards.

Gaskets for joints between manholes, or hand holes and covers shall be as follows, unless otherwise specifically required:

- a. Chemical Tanks..... Rubber
- b. Oil Tanks.....Neoprene
- c. All Other Tanks..... Composition

Gaskets shall be so arranged as to avoid damage to the gasket material when manhole or hand hole covers are removed and replaced.

4.3.3 Instrument nozzles

Instrument nozzles penetrating the lower head of cone or dished-bottom tanks shall be installed in a horizontal position, sloped to drain into the tank, and cut off square not less than 50 mm inside the tank.

4.3.4 Reinforcement

The minimum cross-sectional area of the reinforcement shall be not less than the product of the diameter of the hole, in mm, cut in the shell and the shell-plate thickness.

All reinforcement on the tank shall be designed in accordance with appropriate approved standard.

4.3.5 External Ladders and Platforms

Ladders and platforms shall be attached to the shell by welding or bolting to brackets fabricated from the same material as the shell. The ladder and platform brackets shall be attached to the shell by full-fillet welds.

Ladders and platforms shall be of carbon steel on all tanks.

All ladders and platforms shall be confirmed and approved by the Owner.

4.3.6 Handrails

All handrails shall meet or exceed the applicable requirements of codes and regulatory authorities having jurisdiction.

Handrails shall be of carbon steel.

4.3.7 Supports and Anchors

Supports for raised tanks and anchor plates or anchor rings for flat-bottom tanks shall be provided as required.

Supports shall be capable of being bolted to the floor.

Supports for raised tanks shall be attached to the tank with pads or saddles as required to distribute the load on the tank.

High alloy steel tanks with carbon steel supports shall be provided with a high alloy steel pad of adequate thickness between the dissimilar materials.

4.4 Accessories and Appurtenances

4.4.1 Internal Piping

Internal piping refers to heating coils, cooling coils, air sparger piping and hydraulic mixer piping and bubbler pipe for level instruments.

Internal piping shall be located and supported to provide for necessary expansion and contraction. Supports shall be designed to withstand reaction forces from mixers and forces from maintenance activity by workers in the tank.

Unless specified otherwise, or approved by Owner, all internal piping below the overflow shall be joined by welding and not flanges.

Internal piping shall be bolted to supports to permit easy replacement.

Internal piping support brackets in unlined tanks shall be of the same material as the tank shell. If the tank is lined, the support brackets shall be of stainless steel (if required) or coated to match the tank lining. Carbon steel support brackets shall be minimum of 3/8-inch thick or metric equivalent.

4.4.2 Hydraulic Mixers (if required)

Mixer shall be sized and properly located by Bidder to attain uniform mixing in the minimum time with indicated flow.

Bidder shall be responsible for furnishing a suitable mixer for each application.

4.4.3 Bubbler pipes (if required)

Pipes shall enter through the top of the tank. Penetrations shall be through a 50 mm blind flange on closed-top tanks. The top of the pipe shall terminate 150 mm above the blind flange in a 1/2 NPT thread or metric equivalent.

Pipes shall be vertical throughout their length and not contoured to follow the tank wall.

Pipes shall be loosely supported near the bottom of the tank in a guide bracket to prevent excessive movement during tank agitation but shall be free to be withdrawn from above.

4.4.4 Spray System (if required)

Spray system shall be provided in top of tank and shall be designed to successfully wash the walls of solids with the specified water rate.

4.4.5 Tank Ground

Two electrical pads shall be mounted on each tank located outdoors or not attached to building steel for connection to the ground grid.

4.4.6 Tank Diaphragms (if required)

Tanks shall be equipped with non-permeable, non-flammable membranes. The membranes shall be suitable for the temperature and pressure listed in the Specification.

Diaphragms shall be affixed to the roofs of the tanks with care to prevent localized overstressing and distorting of the diaphragms.

The diaphragms shall be leak-tested with air pressure on the tanks at the design pressure of the tank. Complete manufacturer's literature describing the diaphragms shall be included with the Bidder's proposal.

4.4.7 Air Spargers (if required)

Spargers shall, be designed for uniform mixing of tank contents using the required minimum airflow per surface unit of projected area above the sparger as shall be submitted for review to the Owner.

Spargers shall be located sufficiently, close to the bottom of the tank, by contouring if necessary, to keep solids stirred up.

Air holes shall be a minimum 1/8-inch or metric equivalent diameter to prevent plugging by dirt in the supply air and piping.

Headers shall be enclosed with 60-mesh stainless steel screen to prevent particulates from entering 1/8-inch or metric equivalent air holes.

4.5 Cleaning and Painting

After fabrication and immediately prior to the painting or interior coating process, all the tanks shall be cleaned in accordance with requirement as mentioned in M-001 and S-420.

All welding, machining, cutting operations, hydrostatic testing, non destructive testing and cleaning procedures must be completed prior to the application of interior coating or lining to tanks.

On all work where welding has been done, all surfaces shall have slag and weld spatter removed and shall also be thoroughly cleaned.

Tanks containing corrosive liquids shall be lined with pure hard gum rubber as acceptable to the Owner. Precompounded materials purchased from selfstacks will not be accepted. Lining shall not be less than 5 mm thick throughout. The tank lining shall withstand a liquid at working temperature of the tank. Lining shall be applied by a qualified personnel in accordance with lining manufacturer's recommendation approved by the Owner.

The test shall include spark test to detect pinholes in the coating/lining. The procedure for the spark test shall be submitted to the Owner for approval. Any defects or deficiencies revealed by the tests shall be remedied by the Bidder and such cost shall be borne by him. Other tests as required shall be carried out in accordance with requirement as mentioned in S-420 and S-430.

After fabrication and testing and before flushing out with demineralized water, the interior and exterior surfaces of high alloy steel/corrosion resistant steel tanks shall be thoroughly cleaned to be free of grease, oil, weld spatter, chips or other foreign materials. Visual examination will be made to be sure that no traces of foreign matter remain on the surfaces in contact with the fluid and to give assurance that particulate matters that could damage or foul other system

components are eliminated. After testing, the interior surfaces shall be flushed out with demineralized water. Cleaning shall be done to the complete satisfaction of the Owner.

After cleaning, the outside and inside surfaces of Carbon steel tanks shall be painted in accordance with requirement as mentioned in M-001 and S-420 except otherwise specified.

5 PERFORMANCE TESTING AND GUARANTEES

As a minimum the Bidder shall perform all the tests specified in Clause 3.1 herein.

Testing other than specified herein shall comply with the requirement as mentioned in M-014 "Inspection and Test".

6 QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7.

The quality control documentation shall be as required by the applicable codes.

6.1 Radiography

All radiography shall be in accordance with the procedures and comply to the acceptance standards referenced in AWWA-D100-, latest edition Appendix A.

Spot radiography shall be modified as follows:

One radiograph shall be taken in the first 3 m of weld in each vessel and one spot shall be examined for each additional 1 5 m of full-penetration but welds, or fraction thereof.

At least 2 intersections of vertical and horizontal joints on each tank, when they exist, shall be included in the radiographs.

All films shall be subject to the Owner review and comment.

6.2 Leak Testing (shall be conducted on all tanks):

Each completed tank shall be leak-tested following all fabrication, but prior to painting, coating or lining the interior.

The tank shall be filled with water and inspected frequently during filling. Closed-up tanks for solutions with specific gravities greater than 1 .0 shall be tested by using a standpipe to produce the equivalent pressure on the tank bottom.

The bottom joints of flat-bottomed tanks shall be tested with a vacuum box and soap solution prior to hydrostatic testing.

Heating and cooling pipe coils shall be hydrostatically tested at one and a half times the specified design pressure.

Any leaks detected during hydrostatic testing or vacuum testing shall be repaired by procedures reviewed by the Owner.

High Alloy steel tanks shall be tested with demineralized water or water whose chloride content is less than one part per million.

6.3 Liquid Penetrant Examination:

All liquid penetrant testing shall be per the ASME Code, Section VIII, Appendix VIII, and as approved by the Owner.

All unacceptable defects shall be repaired by procedures reviewed and accepted by the Owner.

6.4 Magnetic Particle Examination:

All welded joints in ferritic material tanks, including nozzle welds, shall be inspected by this method.

All magnetic particle inspection shall be per the ASME Code, Section VIII, Appendix VI, and as approved by the Owner.

All unacceptable defects shall be repaired by procedures reviewed and accepted by the Owner.

6.5 Tank Interior Coating/Lining Testing (applicable to all coated/lined tanks)

The interior coating of tanks shall be tested for integrity after installation and curing, for holidays, pinholes, bonding, etc., by an appropriate test.

The interior coating of tanks shall be tested to determine the thickness of coating applied.

Test procedures for the above shall be submitted to the Owner. No testing work shall be done until these procedures have been approved by the Owner.

Results of all tests shall be documented and submitted to the Owner with a sketch showing areas examined.

Owner reserves the right to witness all tests.

Testing other than specified herein shall comply with the requirements as mentioned in S-420 and S-430.

7 INSTRUMENTATION AND CONTROL

Bidder shall furnish one dial direct reading level indicator where indicated in Clause 3.1. Level indicators shall be complete with float and linkage suitable for operating in the fluid specified.

Where indicated in Clause 3.1 the Bidder shall furnish one 50 mm connection near the top of the tank and one 50 mm connection near the bottom of the tank

The Bidder shall furnish one direct reading dial temperature indicator mounted in a thermowell where indicated in Clause 3.1.

The Bidder shall furnish one bubbler type level indicator where indicated in Clause 3.1.

The Bidder shall furnish one float operated board and cursor type level indicator suitable for reading from a pump and valve control station at a piping manifold where indicated in Clause 3.1.

The Bidder shall furnish gauge glasses where indicated in Clause 3.1 in accordance with Part 4, Technical Specification I-001 "Instrumentation and Control".

The Bidder shall furnish a standpipe for level switches and controllers where indicated in Clause 3.1. The Bidder shall furnish level switches and controllers in accordance with Part 4, Technical Specification I-001 where indicated in Clause 3.1.

8 ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install electrical equipment and accessories required for the Shop Fabricated Tanks. This equipment shall include but not be limited to the following

1. Cathodic Protection System (if required)
2. All electrical control, monitoring and instrumentation devices and other appurtenances required for the Shop Fabricated Tanks, shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Cathodic Protection System

For detailed electrical requirements, see Part 4, Technical Specification E-261.

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